Sessional	I
Program	B. Tech.
Class	3rd Semester
Subject	Strength of Materials-I
Code	MEPC-203
Max Marks	15
Date/Time	24.09.2025/08:00 AM – 08:50 AM

Note: Attempt all the questions. Assume suitably and state, additional data required, if any

1. At a certain point in a piece of elastic material, there are normal stresses of magnitude 120 MPa tensile and 80 MPa compressive acting orthogonally to each other. In addition, there is a shearing stress of 60 MPa acting normal to the normal stresses. Determine
   i. The normal and shear stresses on a plane inclined at 50^0 to the direction of 120 MPa stress.
   ii. The magnitude and direction of principal stresses.
   iii. The magnitude and direction of shear stresses. Also find the normal stress on the plane of maximum shear stress. (5)

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2. An overhanging beam ABCD, 18 m long, is simply supported at A and C 12 m apart. It carries a load of 20 kN concentrated at B which is at a distance of 4 m from A. In addition, there acts a distributed load with intensity varying from zero at A and D to 18 kN/m run at C. Draw the shear force and bending moment diagrams and calculate the position and magnitude of maximum bending moment. Also locate the points of contraflexure, if any. (8)
3. List and give the graphical representation of various theories of failure (2)